

Information Retrieval

Lecture 1

Introduction

Information Retrieval (IR)

- *The indexing and retrieval of textual documents.*
- *Searching for pages on the World Wide Web is the “killer app.”*
- *Concerned firstly with retrieving relevant documents to a query.*
- *Concerned secondly with retrieving from large sets of documents efficiently.*

Typical IR Task

- **Given:**
 - *A corpus of textual natural-language documents.*
 - *A user query in the form of a textual string.*
- **Find:**
 - *A ranked set of documents that are relevant to the query.*

Relevance

- *Relevance is a subjective judgment and may include:*
 - *Being on the proper subject.*
 - *Being timely (recent information).*
 - *Being authoritative (from a trusted source).*
 - *Satisfying the goals of the user and his/her intended use of the information (information need).*

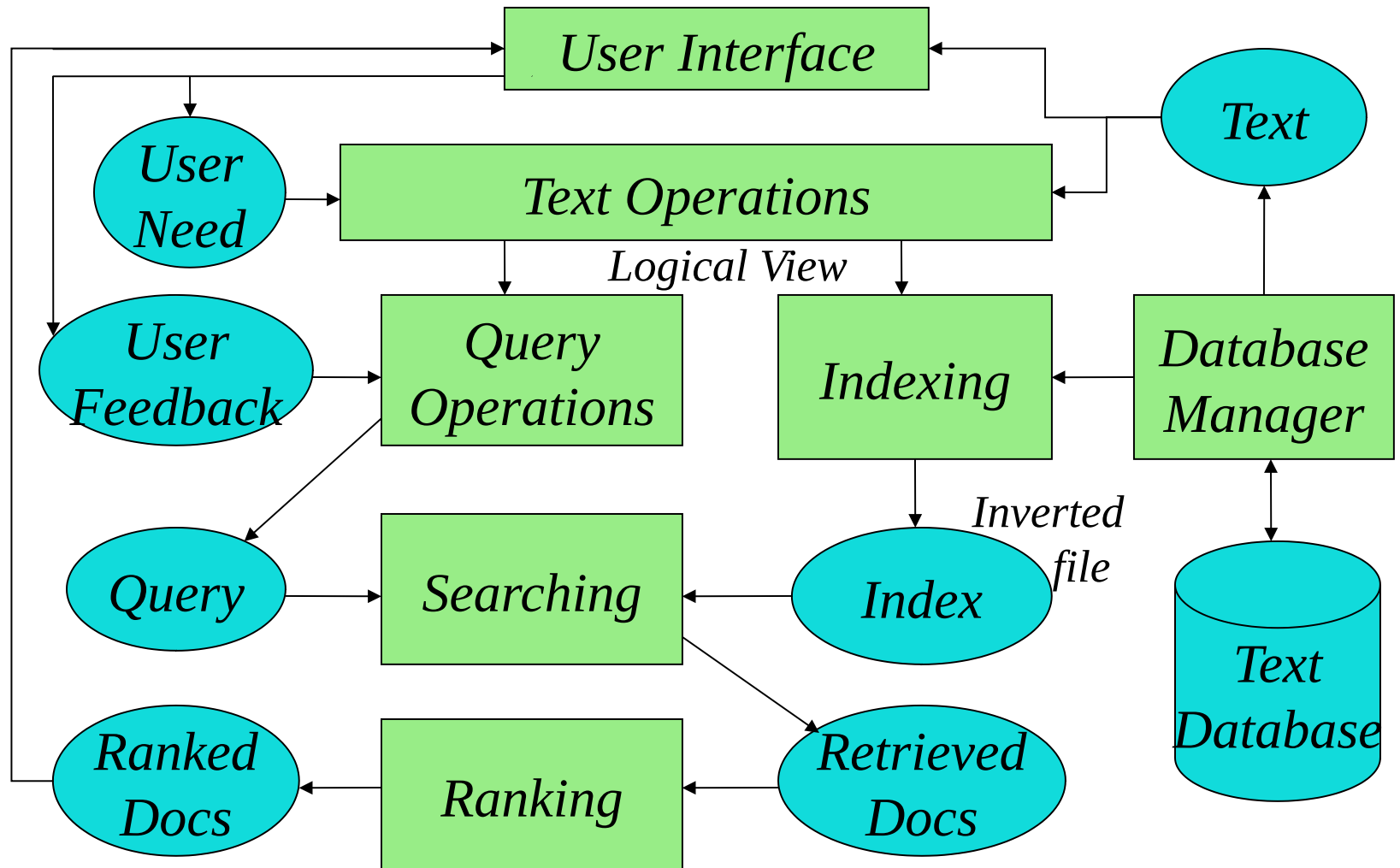
Problems with Keywords

- *May not retrieve relevant documents that include synonymous terms.*
 - “restaurant” vs. “café”
 - “PRC” vs. “China”
- *May retrieve irrelevant documents that include ambiguous terms.*
 - “bat” (baseball vs. mammal)
 - “Apple” (company vs. fruit)
 - “bit” (unit of data vs. act of eating)

Intelligent IR

- *Taking into account the meaning of the words used.*
- *Taking into account the order of words in the query.*
- *Adapting to the user based on direct or indirect feedback.*
- *Taking into account the authority of the source.*

IR System Architecture



IR System Components

- *Text Operations* forms index words (tokens).
 - *Stopword removal*
 - *Stemming*
- *Indexing* constructs an inverted index of word to document pointers.
- *Searching* retrieves documents that contain a given query token from the inverted index.
- *Ranking* scores all retrieved documents according to a relevance metric.

IR System Components (continued)

- *User Interface* manages interaction with the user:
 - Query input and document output.
 - Relevance feedback.
 - Visualization of results.
- *Query Operations* transform the query to improve retrieval:
 - Query expansion using a thesaurus.
 - Query transformation using relevance feedback.

Other IR-Related Tasks

- *Automated document categorization*
- *Information filtering (spam filtering)*
- *Information routing*
- *Automated document clustering*
- *Recommending information or products*
- *Information extraction*
- *Information integration*
- *Question answering*

Related Areas

- *Database Management*
- *Library and Information Science*
- *Artificial Intelligence*
- *Natural Language Processing*
- *Machine Learning*

Database Management

- *Focused on structured data stored in relational tables rather than free-form text.*
- *Focused on efficient processing of well-defined queries in a formal language (SQL).*
- *Clearer semantics for both data and queries.*
- *Recent move towards semi-structured data (XML) brings it closer to IR.*

Library and Information Science

- *Focused on the human user aspects of information retrieval (human-computer interaction, user interface, visualization).*
- *Concerned with effective categorization of human knowledge.*
- *Concerned with citation analysis and bibliometrics (structure of information).*
- *Recent work on digital libraries brings it closer to CS & IR.*

Artificial Intelligence

- *Focused on the representation of knowledge, reasoning, and intelligent action.*
- *Formalisms for representing knowledge and queries:*
 - *First-order Predicate Logic*
 - *Bayesian Networks*
- *Recent work on web ontologies and intelligent information agents brings it closer to IR.*

Natural Language Processing

- *Focused on the syntactic, semantic, and pragmatic analysis of natural language text and discourse.*
- *Ability to analyze syntax (phrase structure) and semantics could allow retrieval based on meaning rather than keywords.*

Natural Language Processing: IR Directions

- *Methods for determining the sense of an ambiguous word based on context (word sense disambiguation).*
- *Methods for identifying specific pieces of information in a document (information extraction).*
- *Methods for answering specific NL questions from document corpora or structured data like FreeBase or Google's Knowledge Graph.*

Machine Learning

- *Focused on the development of computational systems that improve their performance with experience.*
- *Automated classification of examples based on learning concepts from labeled training examples (supervised learning).*
- *Automated methods for clustering unlabeled examples into meaningful groups (unsupervised learning).*

Machine Learning: IR Directions

- *Text Categorization*
 - *Automatic hierarchical classification (Yahoo).*
 - *Adaptive filtering/routing/recommending.*
 - *Automated spam filtering.*
- *Text Clustering*
 - *Clustering of IR query results.*
 - *Automatic formation of hierarchies (Yahoo).*
- *Learning for Information Extraction*
- *Text Mining*
- *Learning to Rank*

Definitions

Information retrieval: Subfield of computer science that deals with automated retrieval of documents (especially text) based on their content and context.

Searching: Seeking for specific information within a body of information. The result of a search is a set of **hits**.

Browsing: Unstructured exploration of a body of information.

Linking: Moving from one item to another following links, such as citations, references, etc.

Definitions (continued)

Query: A string of text, describing the information that the user is seeking. Each word of the query is called a **search term**.

A query can be a single search term, a string of terms, a phrase in natural language, or a stylized expression using special symbols, e.g., a regular expression.

Full text searching: Methods that compare the query with every word in the text, without distinguishing the function of the various words.

Fielded searching: Methods that search on specific bibliographic or structural fields, such as author or title.

Sorting and Ranking Hits

When a **user** submits a **query** to a **search system**, the system returns a set of **hits**. With a large collection of documents, the set of hits maybe very large.

The value to the user often depends on the order in which the hits are presented.

Three main methods:

- **Sorting** the hits, e.g., by date
- **Ranking** the hits by **similarity** between query and document
- **Ranking** the hits by the **importance** of the documents

Information Retrieval from Collections of Textual Documents

Major Categories of Methods

1. Ranking by **similarity to query** (vector space model)
2. Exact **matching** (Boolean)
3. Ranking of matches by **importance of documents** (PageRank)
4. Combination methods

Text Based Information Retrieval

Most **ranking** methods are based on the **vector space model**.

Most **matching** methods are based on **Boolean operators**.

Web search methods combine vector space model with ranking based on importance of documents.

Many practical systems combine features of several approaches.

In the basic form, all approaches treat **words** as **separate tokens** with minimal attempt to interpret them linguistically.

Documents

A textual **document** is a digital object consisting of a sequence of **words** and other symbols, e.g., punctuation.

The individual words and other groups of symbols used for retrieval are known as **tokens** or **terms**.

A textual document can be:

- **Free text**, also known as **unstructured** text, which is a continuous sequence of tokens.
- **Fielded text**, also known as **structured** text, in which the text is broken into sections that are distinguished by tags or other markup, e.g., a library catalog.

[CS/Info 431 covers methods of markup, e.g., XML. Partially structured text, e.g., web pages, is called **semi-structured** text.]

Word Frequency

Observation: Some words are more common than others.

Statistics: Most large collections of unstructured text documents have similar statistical characteristics. These statistics:

- influence the effectiveness and efficiency of data structures used to index documents
- many retrieval models rely on them

Word Frequency

Example

The following example is taken from:

Jamie Callan, *Characteristics of Text*, 1997

Sample of 19 million words

The next slide shows the 50 commonest words in rank order (r), with their frequency (f).

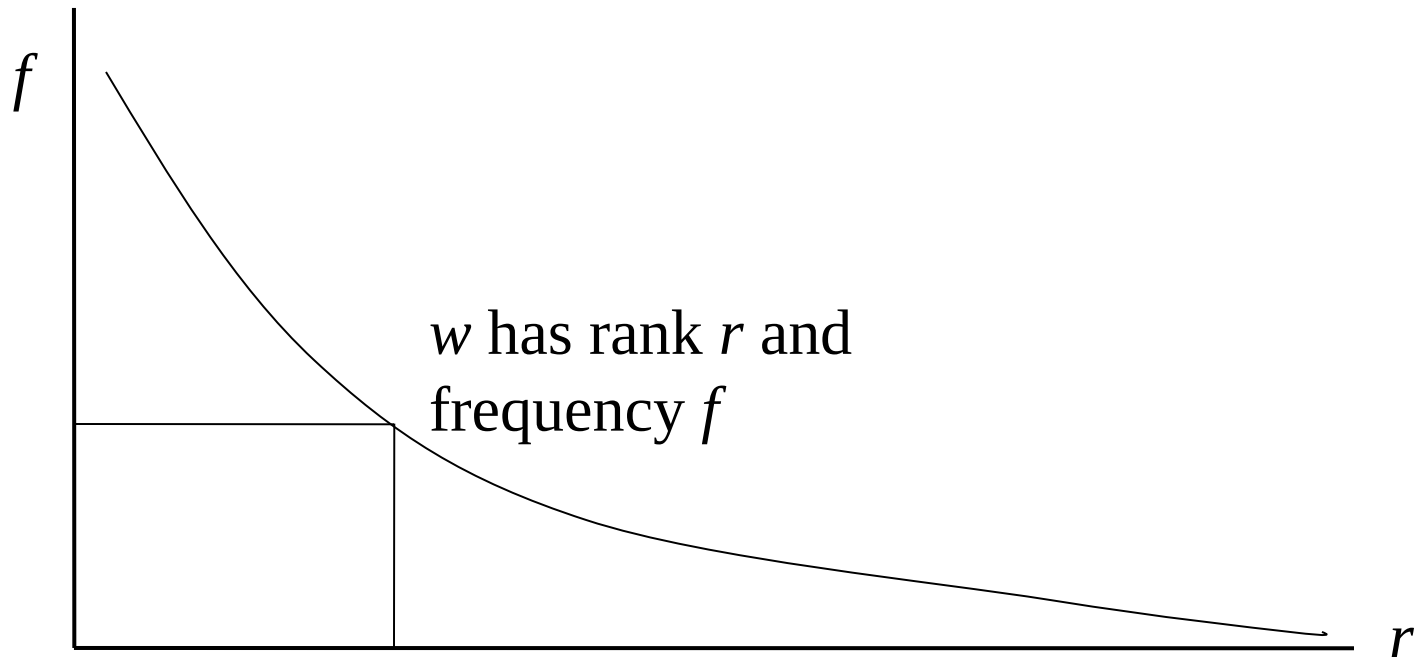
<i>f</i>		<i>f</i>		<i>f</i>	
the	1,130,021	from	96,900	or	54,958
of	547,311	he	94,585	about	53,713
to	516,635	million	93,515	market	52,110
a	464,736	year	90,104	they	51,359
in	390,819	its	86,774	this	50,933
and	387,703	be	85,588	would	50,828
that	204,351	was	83,398	you	49,281
for	199,340	company	83,070	which	48,273
is	152,483	an	76,974	bank	47,940
said	148,302	has	74,405	stock	47,401
it	134,323	are	74,097	trade	47,310
on	121,173	have	73,132	his	47,116
by	118,863	but	71,887	more	46,244
as	109,135	will	71,494	who	42,142
at	101,779	say	66,807	one	41,635
mr	101,679	new	64,456	their	40,910
with	101,210	share	63,925		

Rank Frequency Distribution

For all the words in a collection of documents, for each word w

f is the frequency that w appears

r is rank of w in order of frequency. (The most commonly occurring word has rank 1, etc.)



Rank Frequency Example

The next slide shows the words in Callan's data normalized.
In this example:

r is the rank of word w in the sample.

f is the frequency of word w in the sample.

n is the total number of word occurrences in the sample.

<i>rf*1000/n</i>		<i>rf*1000/n</i>		<i>rf*1000/n</i>	
the	59	from	92	or	101
of	58	he	95	about	102
to	82	million	98	market	101
a	98	year	100	they	103
in	103	its	100	this	105
and	122	be	104	would	107
that	75	was	105	you	106
for	84	company	109	which	107
is	72	an	105	bank	109
said	78	has	106	stock	110
it	78	are	109	trade	112
on	77	have	112	his	114
by	81	but	114	more	114
as	80	will	117	who	106
at	80	say	113	one	107
mr	86	new	112	their	108
with	91	share	114		

Zipf's Law

If the words, w , in a collection are ranked, r , by their frequency, f , they roughly fit the relation:

$$r * f = c$$

Different collections have different constants c .

In English text, c tends to be about $n / 10$, where n is the number of word occurrences in the collection, 19 million in the example.

Zipf's Law

For a weird but wonderful discussion of this and many other examples of naturally occurring rank frequency distributions, see:

Zipf, G. K., *Human Behaviour and the Principle of Least Effort*.
Addison-Wesley, 1949

For a technical understanding of the processes behind this law, take
CS/Info 685, The Structure of Information Networks.

Methods that Build on Zipf's Law

Stop lists: Ignore the most frequent words (upper cut-off). *Used by almost all systems.*

Significant words: Ignore the most frequent and least frequent words (upper and lower cut-off). *Rarely used.*

Term weighting: Give differing weights to terms based on their frequency, with most frequent words weighed less. *Used by almost all ranking methods.*

Definitions

Corpus: A collection of documents that are indexed and searched together.

Word list: The set of all terms that are used in the index for a given corpus (also known as a **vocabulary file**).

With full text searching, the word list is all the terms in the corpus, with **stop words** removed. Related terms may be combined by **stemming**.

Controlled vocabulary: A method of indexing where the word list is fixed. Terms from it are selected to describe each document.

Keywords: A name for the terms in the word list, particularly with controlled vocabulary.

